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Hydrogen hypersonic combined cycle propulsion: Advancements, challenges, and applications

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ABSTRACT

In the progress toward net zero, two propulsion systems are attracting increasing interest, one based on energy stored in batteries, and the other based on energy stored in hydrogen. In between the applications where hydrogen energy storage has more advantages, there is hypersonic propulsion, which is the object of this perspective. Hypersonic vehicles are aircraft or missiles able to travel at speeds above Mach 5. There are four main types of hypersonic vehicles, spaceplanes, glide missiles, cruise missiles, and aircraft. While glide missiles generally do not need propulsion, cruise missiles, and aircraft are designed to sustain hypersonic speeds for extended periods. They use advanced propulsion systems, which include scramjets, relying on the forward motion of the vehicle to compress incoming air for combustion, and featuring supersonic combustion. While scramjets work well at hypersonic speeds, they must be integrated with other propulsion systems, such as ramjets, rockets, and turbojets, to move back and forth from zero speed. Spaceplanes have been designed so far by using only rockets or integrating rockets and turbojets for better operation in the atmosphere. Spaceplanes' operation in the atmosphere could also be further improved by using ram/scramjets. Point-to-point hypersonic travel for civil applications, which is the focus of this work, requires significant advantages in jet propulsion systems that have to work over different regimes, from subsonic to transonic to supersonic to hypersonic. The work reports on the challenges of hydrogen hypersonic combined cycle propulsion.

1. Introduction

Hypersonic vehicles are aircraft or missiles that can travel at speeds greater than five times the speed of sound, or Mach 5. These vehicles are designed to operate in the hypersonic flight regime, which is characterized by extremely high speeds and atmospheric conditions that differ significantly from subsonic or supersonic flight. A historical overview of hypersonic vehicles is offered in Ref. [1]. There are four main types of hypersonic vehicles, spaceplanes, glide missiles, cruise missiles, and aircraft. While glide missiles normally do not need propulsion, cruise missiles, and aircraft are designed to sustain hypersonic speeds for extended periods. Spaceplanes have similar requirements to aircraft and cruise missiles, with the additional need to work in space, where there is no atmosphere. They use advanced propulsion systems, such as scramjets, which rely on the forward motion of the vehicle to compress incoming air for combustion. While scramjets work well at hypersonic speeds, they must be integrated into cruise missiles and aircraft with other propulsion systems, such as ramjets, rockets, and turbojets, to move back and forth from zero speed. Spaceplanes' operation in the atmosphere could also be further improved by using ram/scramjets in addition to rockets and turbojets.

Hypersonic aircraft [2] have the potential to revolutionize long-range travel, as they could significantly reduce flight times for intercontinental travel. Hypersonic cruise missiles [3] which are definitively still challenging but less compared to hypersonic aircraft, may also benefit from the technologies being developed for hypersonic aircraft. Hypersonic glide missiles [4] are then vehicles usually launched using rockets and then approaching the target on descent at hypersonic speeds. Generally, they do not have their propulsion systems but use their high speed and altitude to glide to their target. Hypersonic glide vehicles may also be fitted with scramjets for extended functionalities. Also, hypersonic glide vehicles may ultimately benefit from the development of hypersonic aircraft. Spaceplanes [5] typically use a combination of propulsion systems to achieve their flight and space travel capabilities. The primary propulsion methods employed by spaceplanes are rocket engines. Rocket engines are commonly used in spaceplanes for both atmospheric flight and space travel. Rocket engines work by expelling high-velocity exhaust gases in one direction, creating a

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reaction force that propels the vehicle forward. They are particularly effective in the vacuum of space where there is no surrounding air to provide thrust. Spaceplanes often also rely on jet engines, such as turbofan or turbojet engines, to power spaceplanes during atmospheric flight. These engines work by taking in air, compressing it, and then combusting it with fuel to produce thrust. Jet engines are more efficient than rocket engines at lower altitudes where atmospheric oxygen can be used as an oxidizer, but they become less effective as the air density decreases at higher altitudes. The best spaceplanes employ hybrid propulsion systems that combine elements of both rocket engines and jet engines. These engines typically use a rocket motor for space travel and a jet engine or an air-breathing component for atmospheric flight. This hybrid approach allows for increased efficiency during the atmospheric phase of the flight, as the vehicle can utilize the oxygen in the atmosphere for combustion instead of carrying an oxidizer. Different spaceplane concepts may employ different combinations of propulsion methods to optimize performance and efficiency for their intended purposes.

This work aims to provide a perspective on hypersonic combined cycle (HCC) propulsion [6–9], which represents a cutting-edge area of aerospace engineering and technology, including an overview of HCC propulsion systems, their principles, advantages, challenges, and potential applications, based on a literature review.

2. Hypersonic combined cycle propulsion

Research and development efforts are underway in several countries to advance hypersonic technology for both civilian and military purposes. Here we discuss the use of scramjets as a component of the propulsion system. Since our interest is primarily in civil rather than military operations, we only consider the opportunity of using hydrogen as the fuel, being hydrogen a carbon-free, renewable fuel [10–12]. The use of renewable hydrogen is the avenue to deliver zero–CO_2–emission air transport.

Scramjet, short for Supersonic Combustion Ramjet, is a type of hypersonic propulsion system proposed for high-speed aircraft and missiles [13–18]. Unlike traditional jet engines, scramjets do not have rotating compressor blades. Instead, they rely on the high speed of the incoming air to compress and ignite the fuel. Scramjets operate at extremely high speeds, typically above Mach 5 or hypersonic speeds. The incoming air is compressed as it enters the engine, and fuel is injected into the compressed air stream. The high-speed airflow through the combustor allows for supersonic combustion, where the fuel is rapidly mixed and burned with compressed air. Scramjets are air-breathing engines, meaning they take in atmospheric air to support combustion. This eliminates the need to carry oxidizers, such as liquid oxygen, as in traditional rocket engines. The ability to use atmospheric air as an oxidizer makes scramjets more efficient for long-range high-speed flight in the atmosphere. Scramjets are specifically designed for operation at very high Mach numbers, where the extreme speed is very challenging. The engine's geometry and inlet design are critical to effectively capture and compress the incoming air, maintaining supersonic airflow within the combustion chamber. Even more critical is then to support stable combustion at these speeds under conditions that may change for multiple reasons. Scramjets require a constant supply of fuel to be injected into the airflow for combustion. This typically involves injecting liquid fuel, such as hydrogen, through specialized fuel injectors positioned within the combustion chamber or inlet.

Hypersonic missiles are advanced weapons that travel at extremely high speeds, typically defined as speeds greater than Mach 5. These missiles offer several unique capabilities compared to traditional supersonic or subsonic missiles. Key advantages of hypersonic missiles are speed, maneuverability, range, and warhead options [19–21]. Hypersonic missiles can travel at speeds significantly faster than the speed of sound. This high velocity allows them to reach their targets much more quickly, reducing the time available for defense systems to respond effectively. Hypersonic missiles have the potential for enhanced maneuverability due to their high speed and advanced control systems. This capability enables them to change course, adjust the trajectory, or perform evasive maneuvers to evade interception by anti-missile defenses. Hypersonic missiles have extended range capabilities, enabling them to cover greater distances compared to traditional missiles. This increased range enhances their effectiveness in both offensive and defensive operations. Hypersonic missiles can be equipped with various types of warheads, including conventional explosives, nuclear warheads, or specialized payloads. The specific warhead choice depends on the intended mission, such as precision strikes, anti-ship operations, or strategic deterrence. Hypersonic missiles present significant challenges for existing defense systems due to their high speed and maneuverability. The short reaction time and unpredictable flight paths make it difficult for conventional missile defense systems to intercept them effectively. Hypersonic missile development has prompted a focus on advanced defense technologies to counter this new threat. Developing and deploying hypersonic missiles involves complex engineering and advanced technologies. Achieving and maintaining hypersonic speeds requires advanced propulsion systems and materials capable of withstanding the extreme temperatures and aerodynamic forces experienced during flight. Several countries, including the United States, Russia, China, and others, are actively pursuing hypersonic missile technology. These nations view hypersonic capabilities as a strategic advantage, providing the ability to quickly and accurately strike distant targets while bypassing traditional defense systems. The development and deployment of hypersonic missiles raise concerns about their impact on strategic stability, arms control agreements, and potential destabilization of regional security dynamics.

Several types of hypersonic missiles are being developed by various countries. Examples include Hypersonic Glide Vehicles (HGVs) [22–24], Boost-Glide Missiles [25,26], and Hypersonic Cruise Missiles [27,28]. HGVs are unmanned, maneuverable vehicles that are launched atop a ballistic missile or through other means. Once in space or at high altitudes, they separate from the launch vehicle and glide at hypersonic speeds toward their target. HGVs use their speed, maneuverability, and unpredictable flight paths to evade missile defenses. Boost-glide missiles combine elements of ballistic and cruise missiles. They are launched into space using a rocket booster, after which they separate and glide at hypersonic speeds through the Earth's atmosphere. Boost-glide missiles can maneuver during the glide phase, making them highly unpredictable and difficult to intercept. Hypersonic cruise missiles are powered vehicles that maintain sustained hypersonic speeds throughout their flight. They typically use an air-breathing engine, such as a scramjet, to propel themselves at high velocities. These missiles can be launched from aircraft, ships, or ground-based platforms and can maneuver during their flight to avoid interception.

From a propulsion perspective, hypersonic missiles are designed with a boost-glide or a scramjet-based propulsion system or are gun-launched [19–21]. The first option is intended for much larger ranges, ~1500 (surface-to-surface ballistic or stroke missiles) to ~3500 miles (conventional prompt global strike or non-nuclear intercontinental ballistic missiles), and Mach up to 20. The second option is intended for shorter ranges, ~250 miles, and Mach up to 5–7 (interceptors) or above 8 (strike missiles). Hypersonic cruise or glide missiles fly at much lower altitudes than intercontinental ballistic missiles but can vary much more their trajectories thus resulting to be much less detectable and predictable. Hypersonic missiles gun-launched are then popular in enhanced naval guns and artillery. They are intended for ranges ~100 miles and Mach 6–8. Research and development in the field of hypersonic technology are rapidly evolving, and new concepts and variations of hypersonic missiles are constantly developing. Countries such as the United States, Russia, China, and others are actively engaged in the research, development, and testing of hypersonic missile technology. The pursuit of hypersonic capabilities is driven by the potential advantages they offer in terms of speed, range, maneuverability, and the ability to overcome existing defense systems.

Apart from the launch, glide missiles ultimately enter the glide phase, where they utilize their aerodynamic shape and control surfaces to glide and maneuver toward the target. During the glide phase, no additional propulsion is utilized, and the missile relies on its momentum and aerodynamic lift to maintain its trajectory. However, some advanced glide missiles may also employ scramjet engines, or rocket engines, for propulsion during the glide phase. Scramjet engines operate at high speeds and are capable of efficiently combusting fuel in the air flowing at supersonic speeds. They are designed to intake air at high velocities, compress it, and mix it with fuel for combustion. Scramjets are particularly suitable for hypersonic speeds, offering high efficiency and thrust-to-weight ratio. In a boost-glide configuration, the scramjet engine may be used during the boost phase or the glide phase to provide additional propulsion or sustain high-speed flight.

The propulsion systems used in hypersonic aircraft are specifically designed to provide the necessary thrust and performance required for sustained high-speed flight [29–33]. The design of a hypersonic aircraft is much more challenging than the design of a hypersonic missile. An aircraft needs to take off and land, reach cruise speed and altitude, maintain cruise speed and altitude, and then decrease speed and altitude. This means the propulsion system has to efficiently operate from nominally zero to maximum speed and vice-versa moving through low subsonic, subsonic, transonic, supersonic, and hypersonic range, and this requires multiple propulsion capabilities (for example turbojet engine plus ram/scramjet [29]). If hypersonic missiles are delivery-to-a-point vehicles, hypersonic aircraft are point-to-point vehicles, and this translates into a much more demanding propulsion system.

There are a few examples of hypersonic aircraft. Fig. 1 a presents an artist's concept of an X-51A wave rider (from Ref. [34]). This is a research scramjet hypersonic aircraft. The X-51 completed a flight of about 6 min reaching about Mach 5 for about 3 minutes in 2013. This is the known longest-duration powered hypersonic flight. Wave rider refers to aircraft using compression lift produced by their shock waves. Other developments are not in the public domain. The list of the most popular hypersonic aircraft projects closed or ongoing, then includes REL SKYLON, HSTDV, SR-72, and X-43, Fig. 1 b (from Ref. [35]). While the design of these vehicles is a complex multidisciplinary approach (for example the vehicle aerodynamic is characterized by a complex pattern of shock waves, for example, evidenced by the CFD computation of the flow around the X-43 aircraft working at Mach 7 proposed in Ref. [36]), the focus of this work is the propulsion system.

Scramjets are most efficient at high speeds, but they have a limited operational range at lower speeds due to the requirement for high-speed airflow for compression and combustion. They are primarily intended for hypersonic flight applications where sustained high-speed operation is needed. Fig. 2 a proposes the specific impulse vs. Mach number of hydrogen Turbojet, Ramjet, Scramjet, and Rocket engines, plus the envelope of a typical Rocket Based Combined Cycle (RBCC) and Turbine Based Combined Cycle (TBCC) operation. RBCC combines ram/scramjets with rockets, and TBCC combines turbojet to ram/scramjets, to permit propulsion over subsonic, transonic, supersonic, and hypersonic speeds. Finally, Fig. 2b and c are sketches of RBCC and TBCC propulsion

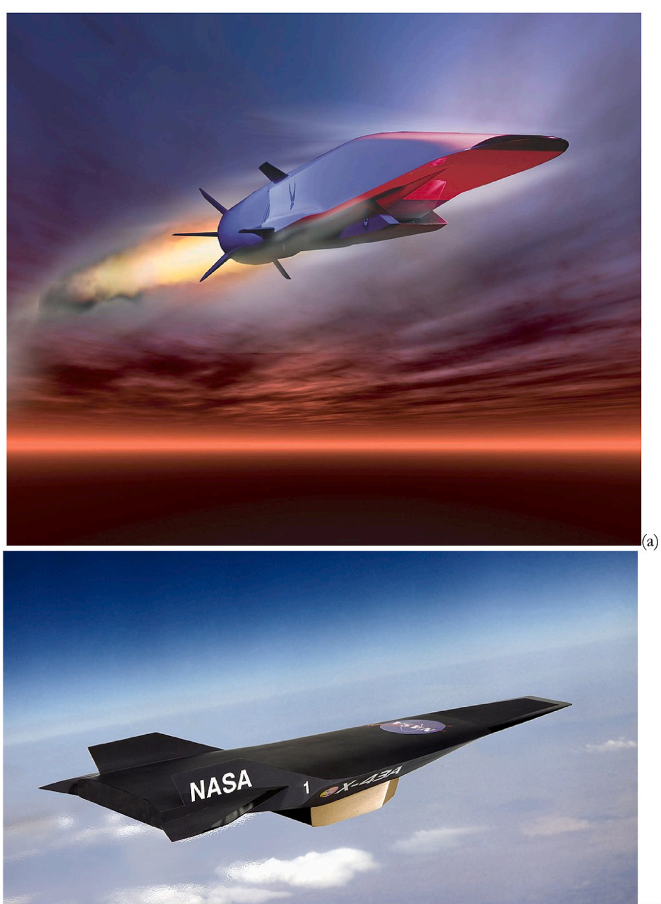

Fig. 1. (a) Artist's impression of an X-51A Waverider. Image from Ref. [34]. Public domain. (b) Artist's impression of an X-43 aircraft. Image from Ref. [35]. Public Domain.

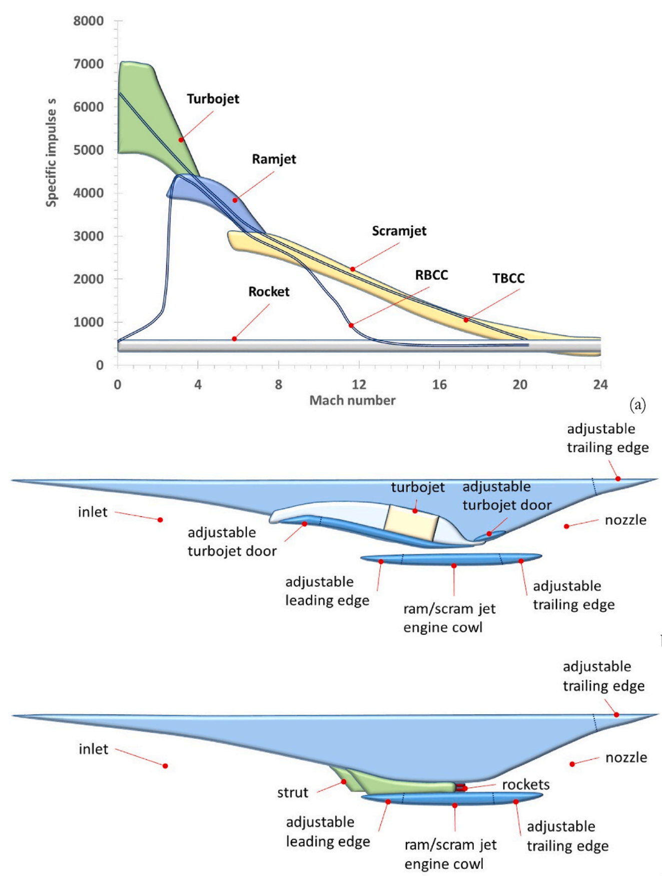

Fig. 2. (a) Specific impulse vs. Mach number of hydrogen Turbojet, Ramjet, Scramjet, and Rocket engines, plus typical RBCC and TBCC operation. Data from Refs. [37,38]. Image reprinted from Ref. [29], Copyright (2022), with permission from Elsevier. (b) TBCC engine comprising turbojet and ramjet/scramjet engines. (c) RBCC engine combining rocket and ramjet/scramjet engines. Image reprinted from Ref. [29], Copyright (2022), with permission from Elsevier.

systems.

As previously mentioned, scramjet propulsion systems are being researched and developed for applications such as hypersonic aircraft, long-range hypersonic missiles, and space launch vehicles, but they are not yet well-established commercial products. They offer the potential for significantly higher speeds and improved efficiency compared to traditional jet engines or rocket propulsion systems. However, there are numerous technical challenges to overcome, such as thermal management, fuel injection and combustion control, and structural integrity at extreme speeds and temperatures. Ongoing research and development efforts aim to advance the capabilities and practical applications of scramjet technology, with significant difficulties in forecasting market uptakes due to the technological hurdles not yet completely solved.

Scramjet engines are commonly part of hypersonic aircraft propulsion. As previously mentioned, scramjets operate at hypersonic speeds and use high-speed airflow to compress and ignite the fuel. They are air-breathing engines that rely on atmospheric oxygen for combustion, eliminating the need to carry the oxidizer on board. Scramjets offer high specific impulse (fuel efficiency) and are designed for sustained supersonic combustion. The inlet design of a hypersonic aircraft is crucial to efficiently capture and compress the incoming air at high speeds. The inlet must ensure that the airflow entering the engine remains at supersonic velocities and provides adequate pressure recovery for combustion. Various inlet designs, such as spike, inward-turning ramps, or dual-mode inlets, are used to achieve efficient air compression and maintain combustion efficiency. Hypersonic flight generates extreme temperatures due to air compression and frictional heating. Thermal management is a critical aspect of hypersonic propulsion to ensure the engine components and structure can withstand high temperatures. Advanced materials and cooling techniques, such as regenerative cooling, are employed to dissipate heat and protect the engine components. Fuel injection in hypersonic propulsion systems must overcome challenges associated with high velocities, air compression, and efficient mixing with the incoming airflow. The fuel is typically injected using specialized injectors designed for supersonic combustion. Hydrogen is a commonly used fuel due to its high combustion efficiency and low molecular weight. Hypersonic propulsion aims to achieve a high specific impulse (Isp), which measures the efficiency of a propulsion system. Specific impulse is directly related to the fuel consumption rate and determines how much thrust can be generated per unit of fuel. High ISP is critical for achieving long-range, efficient hypersonic flight.

Designing a scramjet engine is a complex task that requires expertise in aerodynamics, propulsion systems, materials science, and computational modeling. A simplified overview of the design process for a ram/scramjet engine includes many steps, such as Define Design Requirements, Aerodynamic Design, Hypersonic Flow Analysis, Fuel Injection, Thermal Management, Material Selection, Control and Stability, Integration and System Design, Testing and Validation. Determine the specific goals and requirements of the scramjet engine, such as the desired speed range, thrust levels, and operational conditions. These requirements will guide the design process. Develop the engine's air intake, combustor, and nozzle. The air intake must efficiently compress incoming air and slow it down to subsonic speeds before entering the combustor. The combustor is responsible for mixing and igniting the fuel with compressed air. The nozzle accelerates the exhaust gases to generate thrust. Use computational fluid dynamics (CFD) simulations and wind tunnel testing to analyze the behavior of airflows at hypersonic speeds within the engine components. This helps optimize the design for efficient combustion, minimize drag, and enhance overall performance. Hydrogen provides high-energy content and is suitable for combustion at hypersonic speeds. Determine the injection mechanism and pattern to achieve proper mixing and combustion. Address the extreme heat generated during hypersonic flight by incorporating advanced cooling techniques, such as regenerative cooling or film cooling, to protect the engine components from thermal stresses. Choose materials capable of withstanding high temperatures and pressures encountered during scramjet operation. Materials should possess good thermal stability, strength, and resistance to oxidation and erosion. Design the engine to maintain stable combustion at high speeds and varying conditions. This involves incorporating control mechanisms, such as fuel flow control, throttle control, and feedback systems, to regulate airflow and fuel flow and maintain performance. Integrate the scramjet engine with the overall vehicle or aircraft system, considering factors such as mounting, structural interfaces, fuel storage, and control systems. Compatibility and optimal performance within the complete system is a multidisciplinary task. Perform ground-based testing, including static firing tests, to evaluate the engine's performance, stability, and reliability. This may involve simulating conditions similar to hypersonic flight. Conduct iterative testing and refinement to validate the design.

Supersonic combustion in scramjets poses several technical challenges and issues that need to be addressed for successful engine operation. Key issues associated with supersonic combustion in scramjets include Inlet Design, Fuel-Air Mixing, Flame Stabilization, Heat Transfer, Combustion Efficiency, Ignition, Dynamic Stability, and Computational Modeling. The design of ram/scramjets strongly relies on integrated experiments and simulations [39–46]. The supersonic airflow entering the engine needs to be efficiently slowed down to subsonic speeds for combustion. Designing an effective inlet system that can effectively compress and decelerate the incoming air is a significant challenge. The inlet design should prevent airflow separation, maintain efficient compression, and minimize losses due to shocks and boundary layer interactions. Achieving efficient fuel-air mixing is crucial for effective combustion in scramjets. The extremely short residence time available for combustion in supersonic flow complicates the mixing process. Proper fuel injection techniques, such as transverse injection or cavity-based injection, need to be employed to ensure sufficient fuel-air mixing within the short timescale. Stabilizing the flame in a supersonic flow presents another challenge. The high velocities and complex flow patterns can disrupt the stability of the flame, leading to a flame blowout or flashback. Designing flame holders, such as fuel injectors or cavities, that can maintain flame stability and withstand high-speed flows is critical. Supersonic combustion generates intense heat, and managing heat transfer within the engine is crucial. Thermal management becomes a significant challenge due to the high temperatures encountered during supersonic flight. Developing effective cooling techniques and materials that can withstand the extreme thermal environment is essential to prevent overheating and damage to engine components. Achieving high combustion efficiency is important for optimizing performance and achieving desired thrust levels. Maintaining proper fuel-air ratios, promoting complete combustion, and minimizing losses due to combustion inefficiencies are key considerations in scramjet design. Initiating and maintaining ignition in supersonic flow is challenging due to the high-speed and low-pressure conditions. Ensuring reliable and efficient ignition under these conditions is critical for successful scramjet operation. Scramjets operate in a dynamic and highly unsteady flow environment. The engine needs to be designed to withstand and adapt to changes in flow conditions, including shock waves, boundary layer interactions, and fluctuations in pressure and temperature. Modeling and simulating complex fluid dynamics and combustion processes in supersonic flow is a significant computational challenge. Developing accurate and efficient computational tools for predicting and analyzing supersonic combustion is essential for engine design and optimization. Addressing these issues requires a combination of theoretical analysis, computational modeling, and experimental testing. Advances in computational fluid dynamics, materials science, and combustion research are continuously contributing to the understanding and improvement of supersonic combustion in scramjets.

The design of an engine incorporating a ram/scramjet section is definitively much more complicated than the already difficult design of the ram/scramjet section, and similarly increasingly difficult is the design of a point-to-point, reusable, hypersonic vehicle. Hypersonic aircraft concepts utilize combined engines that incorporate elements of both scramjets and ramjets with turbojets or even rockets [29]. These combined engines, use turbojet mode for subsonic to supersonic flight, and then transition to ramjet, and finally scramjet for sustained hypersonic flight. Alternatively, they may use rockets first, then ramjets, and finally scramjets. This allows for greater operational flexibility and efficiency across different flight regimes.

Turbine based combined cycle (TBCC) typically uses two separate platforms, one a turbojet engine, and one a ram/scramjet, to propel the vehicle. The two separate platforms are typically mounted separately in most of the designs. One first opportunity to combine turbojet and ram/scramjet is offered in Ref. [47]. The propulsion system comprises two separate systems, a low speed, and a high speed. The low speed on top is fed by a low-speed inlet and discharges into the low-speed nozzle, and it comprises a traditional turbojet engine for a high Mach number. The high speed on the bottom is a ram/scramjet engine with a variable geometry inlet and a variable geometry nozzle. In Ref. [18], the variable inlet is similarly split in two. The upper part of the inlet feeds an integrated turbojet/ramjet, or turboramjet, which works for Mach 0–4. The lower part of the inlet is then used to feed a dual mode ram/scramjet, which works for Mach >4, namely a ramjet (Mach 4–6) with subsonic combustion and a scramjet (Mach 6–10) with supersonic combustion. Better integration is proposed in the Unified Turbine Based Combined Cycle (UTBCC) of [48], which is a propulsion system that couples a turbojet engine and a dual-mode ramjet/scramjet into a single propulsion platform [29]. While this is certainly more challenging, it is a much better design once properly developed. The practical definition of the ram/scramjet flow path includes the geometry of the inlet, isolator, combustor, and nozzle sections.

Developing practical and reliable propulsion systems for hypersonic aircraft is a complex engineering challenge. Research and development efforts continue to focus on improving fuel efficiency, thermal management, structural integrity, and overall performance to enable the realization of hypersonic flight capabilities for various applications, including long-range transportation, space access, but also military applications.

The commercial perspective of propulsion by scramjet engines holds potential for various applications and industries, despite different analysts having different opinions [29,49–53]. Some key points to consider include Hypersonic Travel, Space Launch Systems, Military Applications, Research and Development, Economic Growth and Job Creation, and Technological Spin-Offs.

Scramjet engines can propel vehicles at hypersonic speeds, which could revolutionize air travel. With hypersonic propulsion, long-distance flights could be completed in significantly reduced time compared to traditional subsonic aircraft. This could lead to a paradigm shift in commercial aviation, enabling faster and more efficient global transportation. Scramjet engines can also be utilized in space launch systems, specifically for the atmospheric phase of the launch. By using air-breathing engines like scramjets during the initial part of the ascent, the need for carrying oxidizers (such as liquid oxygen) onboard can be eliminated, reducing the overall launch mass and cost. This can potentially make access to space more affordable and enable the launch of larger payloads. Scramjet technology has significant military applications. High-speed scramjet-powered missiles or hypersonic glide vehicles can provide enhanced capabilities for defense purposes. They can deliver rapid response times, maneuverability, and increased range compared to traditional missile systems, offering a strategic advantage in various scenarios. The development of scramjet engines for commercial purposes drives research and development efforts in aerospace and propulsion technologies. This leads to advancements in materials science, aerodynamics, combustion, and thermal management, which have wider applications beyond scramjet engines. The knowledge and technology developed for scramjets can be leveraged in other fields, contributing to technological progress and innovation. The commercialization of scramjet technology can foster economic growth and job creation. It stimulates investment in research, development, and manufacturing infrastructure. It also creates employment opportunities across a range of industries, including aerospace, engineering, manufacturing, and associated support services. The development of scramjet engines may lead to technological spin-offs that can benefit other sectors. Advances in high-temperature materials, aerodynamics, combustion processes, and thermal management techniques have applications beyond scramjets. These advancements can find utility in industries such as energy, automotive, and environmental sectors. The commercialization of scramjet engines is a complex and multi-faceted process. It requires significant investments, research, and regulatory considerations. Challenges related to cost, safety, scalability, and infrastructure need to be addressed for widespread commercial adoption. Furthermore, the successful integration of scramjet technology into commercial applications depends on factors such as market demand, regulatory approvals, environmental considerations, and public acceptance. Collaboration between industry, government agencies, and research institutions is essential for realizing the full commercial potential of scramjet engines. About 70 years after the first proposal of supersonic combustion ramjets [36], the success of scramjets is likely approaching, hopefully, to deliver better civil transport, faster and at a reduced environmental and economic cost, rather than produce new weapons.

3. Conclusions

As we strive for net-zero progress, two propulsion systems are gaining attention. One system is relying on battery energy storage, and the other on hydrogen energy storage. Within applications favoring hydrogen energy storage, hypersonic propulsion stands out with distinct advantages. This work has provided a perspective on hypersonic combined cycle (HCC) propulsion based on public domain information and published works. HCC represents a cutting-edge area of aerospace engineering and technology, with advantages, challenges, and potential applications, which also include defense, making the subject extremely sensitive. HCC propulsion differs from traditional rocket and air-breathing engines. The key components of an HCC engine, include scramjets, ramjets, and rockets. Advantages of HCC propulsion include its efficiency across a wide speed range. Major challenges associated with developing and operating HCC propulsion systems include heat management, engine integration, and material requirements. HCC engines have a growing role in space access, global travel, and defense systems. HCC engines could revolutionize space exploration, travel, and defense. HCC propulsion is an important step in advancing hypersonic technology.

Notes

Views expressed in this Viewpoint are those of the author and not necessarily the views of the journal.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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